CLASS 10 MATHEMATICS :: FA-3

Generating New Problems

Student Name: ..... **Roll No:** ..... **Class & Sec:** .....

1. Find the distance between the coordinate points A(2, 3) and B(6, 6) using the distance formula.

2. Determine the coordinates of the midpoint of the line segment joining P(−2, 8) and Q(6, −4).

3. Find the area of the triangle whose vertices are given by (1, 1), (4, 1), and (4, 5).

4. In ΔABC, DE || BC. If AD = 2 cm, DB = 3 cm, and AE = 4 cm, calculate the exact length of EC.

5. The perimeters of two similar triangles ΔABC and ΔPQR are 36 cm and 24 cm respectively. If PQ = 10 cm, find length AB.

6. A vertical stick 12 m long casts a shadow 8 m long on the ground. At the same time, a tower casts a shadow 40 m long. Find the height of the tower.

7. A point P is 13 cm away from the centre of a circle of radius 5 cm. Calculate the length of the tangent drawn from point P to the circle.

8. If tangents PA and PB from an external point P to a circle with centre O are inclined to each other at an angle of 80°, find the measure of ∠POA.

9. Calculate the area of a sector of a circle with radius 6 cm if the sector angle θ = 60°. (Take π = 22/7)

10. Find the length of an arc of a circle of radius 21 cm that subtends an angle of 60° at the centre.